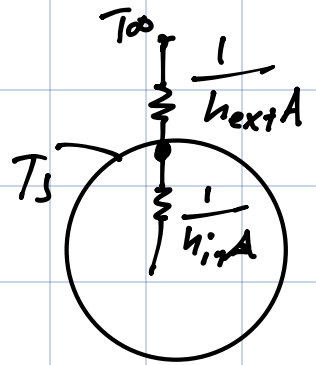
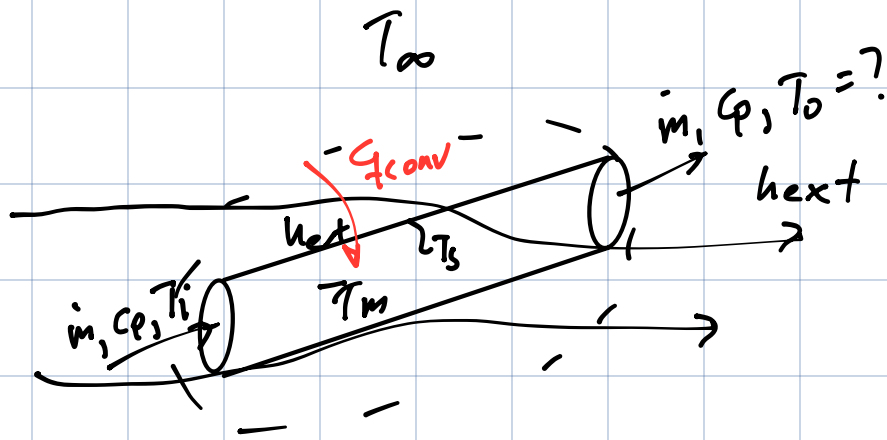
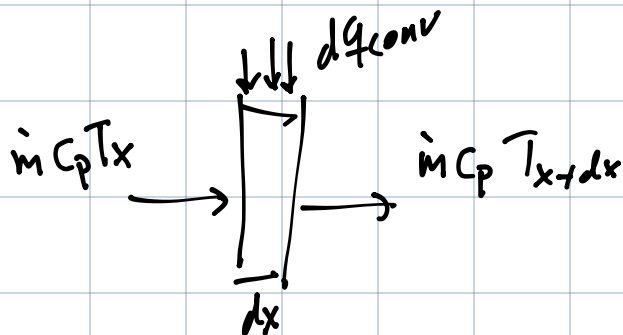
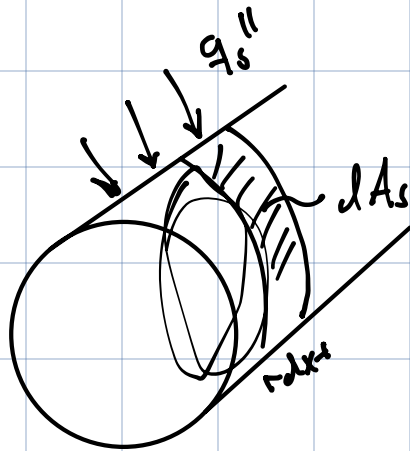




$$q_{conv} \approx h_{in} \pi D L (\bar{T}_s - \bar{T}_m)$$

$$q_{conv} = \dot{m} C_p (\bar{T}_o - T_i)$$

$$h_{ext} \pi D L (\bar{T}_\infty - \bar{T}_s) = \dot{m} C_p (\bar{T}_o - T_i)$$



$$\dot{m} C_p (T_x - T_{x+dx}) + q_s'' \pi D dx = 0$$

$$= \dot{m} C_p dT + q_s'' \pi D dx = 0$$

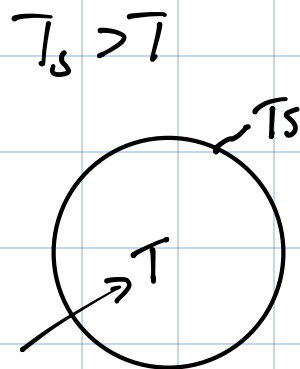
$$\dot{m} C_p \frac{dT}{dx} = q_s'' \pi D$$

$$\frac{dT}{dx} = q_s'' \frac{\pi D_i}{\dot{m} C_p}$$

$$T = \frac{q_s'' \pi D_i}{\dot{m} C_p} x + C_1$$

$$x=0 \quad T=T_{in}$$

$$T = \frac{q_s'' \pi D_i}{\dot{m} C_p} x + T_{in}$$



$$\dot{m} C_p (T_x - T_{x+dx}) + h_i \pi D_i dx (T_s - T) = 0$$

$$-\dot{m} C_p \frac{dT}{dx} + h_i \pi D_i (T_s - T) = 0$$

$$\frac{dT}{dx} = \frac{h_i \pi D_i}{\dot{m} C_p} (T_s - T)$$

$$\Delta T = T_s - T$$

$$d\Delta T = -dT$$

$$-\frac{d\Delta T}{dx} = \frac{h_i \pi D_i}{\dot{m} C_p} \Delta T$$

$$\int_i^0 -\frac{d\Delta T}{\Delta T} = \int_i^0 \frac{h_i \pi D_i}{\dot{m} C_p} dx$$

$$-\ln \left(\frac{\Delta T_0}{\Delta T_i} \right) = \frac{h_i \pi D_i L}{\dot{m} C_p}$$

$$\checkmark \frac{T_s - T_0}{T_s - T_i} = \Delta T_0 = \Delta T_i e^{-\frac{h_i \pi D_i L}{\dot{m} C_p}}$$

$$\dot{m} = \rho V A$$

$$V = \frac{\dot{m}}{\rho A}$$

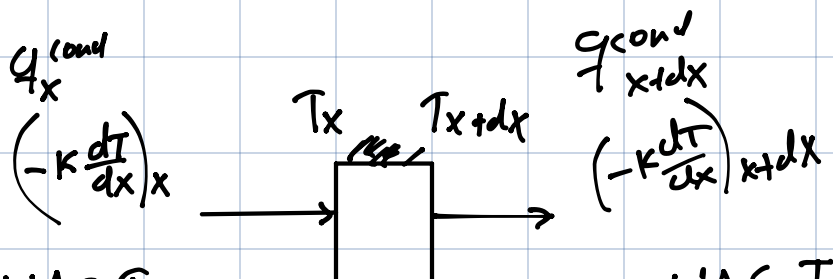
$$Re_D = \frac{\rho D V}{\mu}$$

$$= \frac{\phi D \dot{m}}{\rho A \mu}$$

$$= \frac{\dot{m} D}{\mu A}$$

$$= \frac{\dot{m} D}{\mu \frac{\pi D^2}{4}}$$

$$Re_D = \frac{4 \dot{m}}{\pi D \mu}$$



$$\rho U A C_p T_x \longrightarrow \boxed{\hspace{1cm}} \longrightarrow \rho U A C_p T_{x+dx}$$

dx

$$\dot{q}_x^{cond} - \dot{q}_{x+dx}^{cond} + \rho U A C_p T_x - \rho U A C_p T_{x+dx} = 0$$

$$-\frac{d\dot{q}^{cond}}{dx} dx - \rho U A C_p dT = 0$$

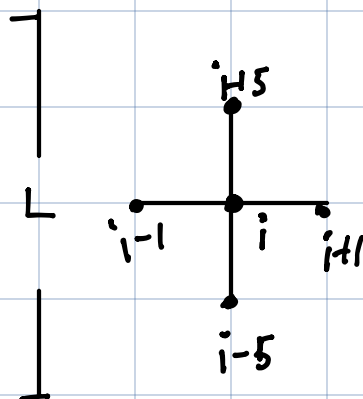
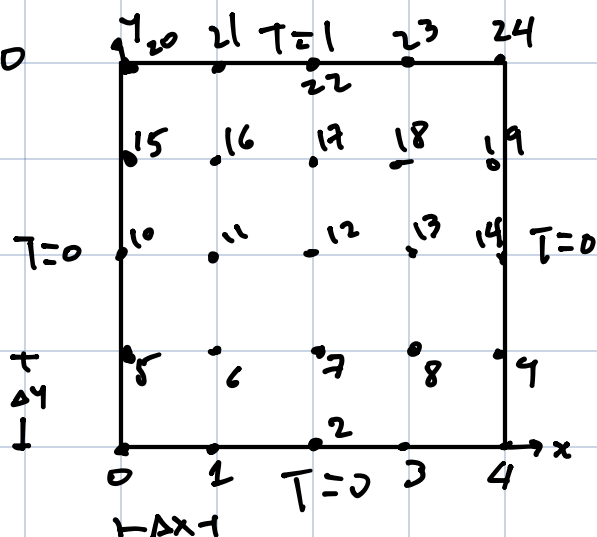
$$-\frac{d}{dx} \left(-k A \frac{dT}{dx} \right) dx - \rho U A C_p dT = 0$$

$$\frac{d}{dx} \left(k A \frac{dT}{dx} \right) - \rho U A C_p \frac{dT}{dx} = 0$$

$$U \frac{dT}{dx}$$

$$\vec{U} \cdot \nabla T$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$



$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2}$$

$$\frac{\partial^2 T}{\partial y^2} = \frac{T_{i+5} - 2T_i + T_{i-5}}{\Delta y^2}$$

$$\frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2} + \frac{T_{i+5} - 2T_i + T_{i-5}}{\Delta y^2} = 0$$

$i = 6, 7, 8,$
 $11, 12, 13,$
 $16, 17, 18$

$$i = 0, 1, 2, 3, 4 \quad T_i = 0$$

$$i = 5, 10, 15, \quad T_i = 0$$

$$i = 20, 21, 22, 23, 24 \quad T_i = 1$$

$$i = 9, 14, 19 \quad T_i = 0$$

$$\begin{matrix}
 & T_0 & T_1 & T_2 & T_3 & \dots & T_5 & T_6 & T_{20} & & T_{24} \\
 \begin{matrix} i=0 \\ i=1 \\ i=2 \\ i=3 \\ \vdots \\ i=6 \\ i=20 \\ \vdots \\ i=24 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \dots & 1 & \dots & 0 \\ 0 & 1 & 0 & 0 & 0 & \dots & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \frac{1}{\Delta y^2} & 0 & 0 & \dots & \frac{1}{\Delta x^2} & -\frac{2}{\Delta x^2} \frac{1}{\Delta y^2} & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & \dots & 1 & 0 & 0 \end{bmatrix} & \times & \begin{bmatrix} T_0 \\ T_1 \\ T_2 \\ T_3 \\ \vdots \\ T_{24} \end{bmatrix} & = & \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \end{bmatrix}
 \end{matrix}$$